Label-Only Black-Box Adversarial Attack on a CIFAR-10-Style CNN with Minimum-L_2 Noise on the 8-Bit Pixel Grid

Santhosh Kandagatla and Bhargav Reddymalla
Team Friday
Dept. of CSE (Cybersecurity), Geethanjali College of Engineering and Technology
Hyderabad, India

***Abstract* — We attack a pretrained CIFAR-10-style CNN in the strict hard-label black-box setting, where only the predicted class of a queried image is observable, and produce one adversarial PNG for each of 1000 clean images with as little visible noise as possible. The attack is a HopSkipJump-style decision-based method: start points come from nearby other-class images, the boundary normal is estimated from label-only probes, and a geometric step followed by binary search shrinks the perturbation. A final stage works on the 8-bit grid, so the saved files and not only their floating-point versions are adversarial, and two restarts keep the smaller valid perturbation per image. Measured on reloaded PNG files, the attack fools the model on 1000/1000 images at every tested batch size, with mean L_2 of 0.2073 on the [0,1] pixel scale (0.833 on normalized inputs), mean PSNR of 50.5 dB, mean SSIM of 0.998 and 18,455 queries per image. A pixel-pruning refinement lowers L_2 by a further 4.4%, but about half of the pruned images flip back under input noise of 10^{-4}, so we reject it. The imperceptibility metric is unspecified, so we report several.**

***Index Terms* — adversarial examples, black-box attack, decision-based attack, HopSkipJump, imperceptibility, image classification**

I. Introduction

Deep classifiers can be fooled by small input perturbations [1], [2]. In the black-box setting the attacker may query the model but cannot inspect its architecture, parameters, intermediate outputs or class scores. The hardest variant exposes only the predicted label, which rules out gradient-based and score-based attacks and calls for decision-based methods [3], [4].

This report covers Problem 2 of a challenge: for each of 1000 clean images, produce one adversarial PNG that changes the model's prediction. Entries are scored on attack success rate and on the imperceptibility of the noise, with equal weight; the number of model queries and the change in prediction confidence only break exact ties. We therefore treat queries as a budget to spend on lowering the noise, while insisting that success holds for the *saved files*.

Our contributions are (i) a label-only attack that works on the 8-bit PNG grid and is verified on every saved file; (ii) a restart scheme that lowers mean L_2 from 0.2190 to 0.2073; and (iii) a robustness-aware evaluation, including the reason we do *not* submit a version with 4.4% less noise.

II. Data and Problem Setting

The data are 1000 clean 32×32 RGB images in the style of CIFAR-10 [5], 100 for each of ten classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck), and a pretrained TorchScript CNN that returns one label per input. Inputs are normalized with a fixed per-channel mean and standard deviation. The model classifies all 1000 clean images correctly. It is called only as `model(x)`.

Pixels are 8-bit, so a perturbation is a set of integer changes in $0..255$. For a clean image x with prediction c we want an adversarial image x' with $f(x') \neq c$ that minimizes $\|x' - x\|_2$ on the [0,1] scale. The organizers do not name the imperceptibility metric, so we optimize L_2 and also report L_∞, PSNR and SSIM.

III. Method

A. Start points

A decision-based attack needs one image that already fools the model. For each image we rank the clean images of the other classes by pixel L_2 distance and take the 10 nearest. Each candidate that the model labels differently is pulled toward x by a 12-step binary search along the connecting segment, and the closest result is kept. Only pixel distances and returned labels are used.

B. Boundary refinement

At iteration t the current boundary point x_t lies at distance d_t from x. We draw $B_t = \min(\lfloor 50\sqrt{t} \rfloor, 800)$ random unit directions u_b, query $x_t + \delta_t u_b$ with $\delta_t = \sqrt{D}\,\theta\,d_t$ ($D = 3072$, $\theta = 10^{-3}$), and set $\phi_b = +1$ if the label differs from c and -1 otherwise. The boundary normal is estimated as a baseline-subtracted Monte-Carlo average [3]:

$$\hat{g}_t = \sum_{b=1}^{B_t} \big(\phi_b - \bar{\phi}\big)\, u_b, \qquad \bar{\phi} = \tfrac{1}{B_t} \sum_b \phi_b, \tag{1}$$

normalized to unit length (if all probes agree, the raw signs are used). A step $x_t + \epsilon_t \hat{g}_t$ with $\epsilon_t = d_t/\sqrt{t}$ is tried first and halved up to ten times until it is adversarial. A binary search back toward x then recovers distance, and the new point is kept only if it is closer than x_t. Probes and the search run in floating point, which keeps small probe radii informative. We run 40 iterations.

C. PNG-safe finalization

The refined point is rounded to the 8-bit grid. Some images sit so close to the boundary that rounding removes the effect; for those the perturbation is scaled by 1.2 repeatedly (at most 40 times) until the rounded image is adversarial. A final 10-step binary search with rounding inside the loop returns the smallest scaling that survives saving.

D. Restarts

The attack runs twice with seeds 0 and 1. The second run draws its 10 start candidates at random from the 30 nearest other-class images, and for each image the valid result with the smaller L_2 is kept:

$$x^\star = \arg\min_{r\in\{1,2\}} \|x^{(r)} - x\|_2 \quad \text{s.t. } f(x^{(r)}) \neq c. \tag{2}$$

IV. Evaluation

A. Protocol

All numbers are measured on the 1000 saved PNG files, reloaded from disk with the setup cell's own transform. An image counts as a success if its predicted label differs from the clean prediction; since all clean images are classified correctly, this equals misclassification. Success was also checked at batch sizes 1000, 250, 32 and 1 through the same `model(x)` call. The query count includes the setup cell's clean query and every query made by the attack. The attack is implemented in PyTorch [6] and the full run took 6295 s (≈105 min) on a Colab GPU. SSIM [7] uses an 11×11 Gaussian window (σ=1.5) per channel on [0,1] images, and PSNR is computed per image.

B. Results

Table I summarizes the submitted set. The smallest perturbation changes a single value by one level ($L_2 = 1/255 \approx 0.0039$), the floor of the

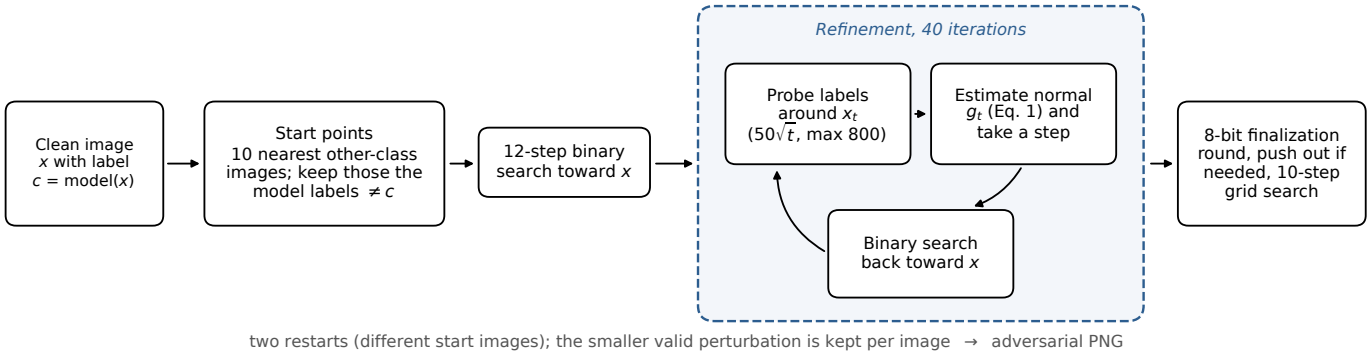


Fig. 1. Attack pipeline. Every model call returns only a label. Start points come from other-class images, the boundary normal is estimated from label-only probes (Eq. 1), and the last stage works on the 8-bit grid so the saved PNG is itself adversarial.

TABLE I
FINAL RESULTS ON THE SAVED PNG FILES

Metric	Value
Attack success rate	100% (1000/1000)
Success at batch sizes 1000/250/32/1	1000/1000 each
Mean L_2 (pixel $[0, 1]$ / normalized)	0.2073 / 0.8332
L_2 median / 90th / 99th pct	0.192 / 0.376 / 0.565
L_2 max / min	0.911 / 0.0039
Mean L_∞ (0–255); median / 90th / 99th / max	4.95; 5 / 9 / 14 / 20
Images with $L_\infty \leq 8$ / ≤ 10	85.1% / 93.1%
Mean PSNR (worst image)	50.53 dB (35.68 dB)
Mean SSIM (worst image)	0.9980 (0.9830)
Mean absolute change per value	0.645 / 255
Queries per image (total)	18,455 (18.5 M)

TABLE II
NOISE VERSUS EFFORT AND VARIANTS

Version	Mean L_2 (px / norm.)	PSNR (dB)	Queries
Single run, 25 it.	0.2717 / 1.090	48.35	4,741
One restart, 40 it.	0.2190 / 0.880	—	9,228
Submitted (2 restarts)	**0.2073 / 0.833**	**50.53**	**18,455**
Submitted + pruning (rejected)	0.1981 / 0.797	50.97	≈19.4k

format. About 46% of the 3072 values change, and the mean absolute change over all values is 0.65 levels out of 255. Fig. 2 shows examples; the noise is not visible in the adversarial images themselves.

C. Where the noise goes

The per-image noise is right-skewed (Fig. 3): the median is 0.192 against a mean of 0.207, the 99th percentile is 0.565 and the maximum 0.911. Classes differ by almost a factor of two (Fig. 4). Truck and automobile images need the most noise (0.282 and 0.273), dog, airplane and bird the least (about 0.16). We did not investigate why; a plausible but untested explanation is that the vehicle classes sit farther from their nearest decision boundary.

D. Effect of design choices

Table II and Fig. 5 show the effort/noise trade-off. The second restart lowers L_2 by 5.3% over one restart, and the full configuration by 23.7% over our first one. Returns diminish: doubling the queries buys 5.3%.

Two design choices matter more than the query budget. First, probes must not be rounded to the 8-bit grid: in a spot check on one image with 20 probes, every probe at $L_2 = 0.2$ rounded back onto the unperturbed point (80% at 0.5, none at 1.0), so the gradient estimate carries no information exactly when fine gains happen. Second, shrinking the step after *successful* candidates and stopping at the first failure leaves each step tiny. An early variant with both problems barely improved on its start points (mean L_2 3.85 on a 20-image test, against 0.2073 now).

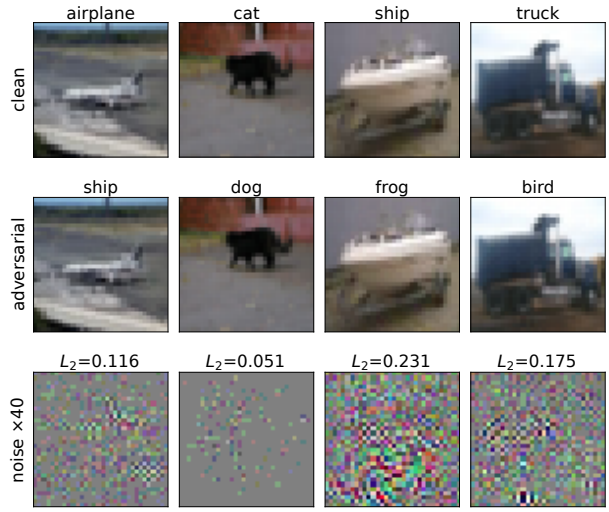


Fig. 2. Clean image, adversarial image and the noise amplified $40\times$ (gray = no change). Titles give the model's predicted class and the image's L_2.

E. Robustness and the pruning trade-off

A minimum-noise attack ends close to the decision boundary, so we measured how stable the saved images are. Pixel pruning moves groups of changed values one level back toward the clean image and keeps a change only if the image stays adversarial. It lowered mean L_2 by 4.4% and passed our checks in the Colab run (100% success, including one image at a time on CPU). Margin tests tell a different story (Table III): input noise of 10^{-4} in normalized units flips 499 of 1000 pruned images back to the clean label (in at least one of three random draws) against 14 of the submitted ones, and one pruned image already fails in ordinary float32 at batch size 32. A hybrid that keeps pruning only where an image survived the noise test cut L_2 just 2.7% (0.2016) and still flipped 59 of 200 images in float16.

Because success is half of the score and a 4.4% noise change is far below visibility, we submit the unpruned set. The float64 and reduced-precision tests were run offline on a copy of the model purely as diagnostics; they are not part of the attack, which only calls `model(x)`.

V. LIMITATIONS

- **Unknown metric.** We minimize L_2. A pure L_∞ ranking could favor flatter noise; our mean L_∞ is 4.95 levels (max 20).
- **Boundary-hugging images.** The confidence change is small by construction, and under a deliberately harsh float16 test 4 of 200 sampled images flip back.
- **High query cost.** About 18.5k queries per image, spent on lower

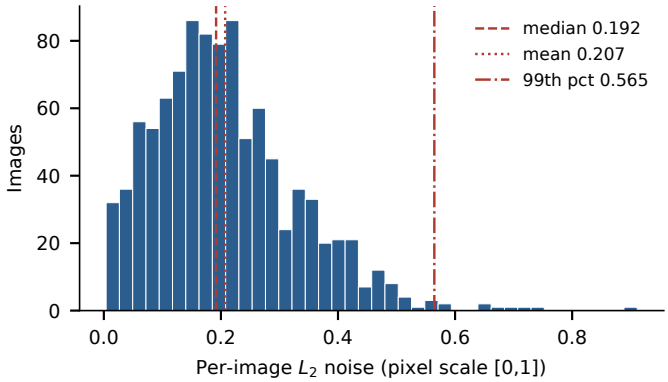


Fig. 3. Per-image L_2 noise over the 1000 saved images.

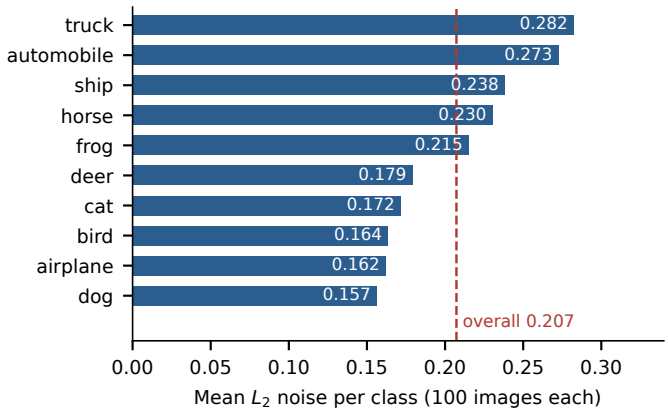


Fig. 4. Mean L_2 noise by class.

noise because queries only break exact ties.

- **Start points from the provided set.** The rules do not mention using other clean images as start points; the attack needs only their pixels and the model's labels, and we state the choice explicitly.
- **Narrow robustness evidence.** Reduced-precision tests used 200 sampled images on one machine, and we have not seen the organizers' evaluation setup.

VI. Conclusion

A label-only decision-based attack with float-space refinement, a quantization-aware final stage and two restarts yields 100% success on 1000 verified PNGs with mean L_2 of 0.2073, PSNR of 50.5 dB and SSIM of 0.998. The remaining gains, such as pixel pruning, cost robustness, so we stop here. The attack is seeded, and two full runs gave byte-identical images. The notebook, the 1000 adversarial PNGs and the README are in the accompanying repository.

References

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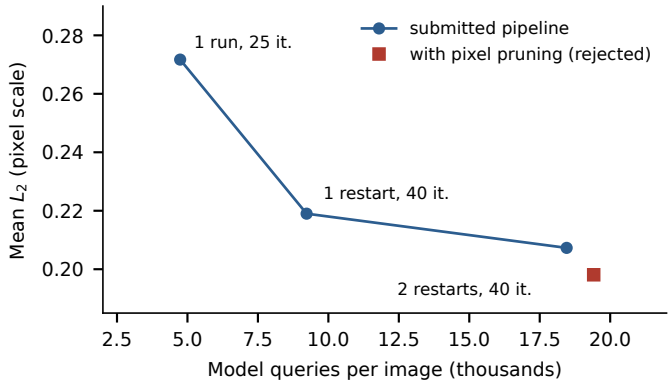


Fig. 5. Mean L_2 against queries per image.

TABLE III
Margin Tests: Images Flipping Back to the Clean Label

Test	Submitted	Pruned
Input noise $\sigma{=}10^{-4}$ (of 1000)	14	499
Input noise $\sigma{=}10^{-3}$ (of 1000)	152	874
float16 (200 sampled)	4	102
bfloat16 (200 sampled)	23	85
float32/float64 CPU, several batch sizes (of 1000)	0	1

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